

(a) Patch and array extent

```
M1 // N := MAX_PIPES * 2
M2 struct dc_link *links[N];
M3 // get_host_router_...(...)
M4 ...
M5 - for (i=0; i<N; ++i) {
M6 + for (i=0; i<N-1; ++i) {
M7 if (!dc->links[i] || ...)
M8 continue; ...
M9 if (hr_match) {
M10 primary=dc->links[i];
M11 secondary=dc->links[i+1];
M12 ...;
M13 break;
M14 }
M15 }
```

Vulnerable: $\max(i+1) = N$

Fixed: $\max(i+1) = N - 1$

① extent N

② $B = N$

④ $B = N - 1$

③ $i + 1$

(b) Starting checker

B1 checkASTCodeBody(F) :

B2 for loop $i < B$:

B3 find access $A[i+1]$

B4 if no body guard:

B5 report($A[i+1]$)

⑤ Missing extent test; V/F: 1/1

(c) Automatic refinement

C1 checkASTCodeBody(F) :

C2 keep loop/access discovery

C3 $N = \text{type_extent}(A)$

C4 $\text{max_index} = (B-1)+1$

C5 if $\text{max_index} < N$:

C6 suppress; else: warn

Tested guards normalized; no name filter

⑥ Extent-aware condition; V/F: 1/0

Solid: bound-to-access links (orange: vulnerable; green: fixed). Dashed: correspondences.